EXP NO: 12

Wireshark packet capture and Analysis

Aim:

To capture and analyze network packets using wireshark and study different network protocols

Software Required:

wireshark and windows OS

Procedure

1. Open wireshark and select the active network interface
2. Start capturing packets and approximately 100 packets
3. Save the capture file in .pcapng format
4. Apply display filters to analyze different protocols
5. Inspect packet details and generate flow graphs where required

Packet Analysis

a) TCP and UDP

tcp || udp

TCP provides reliable communication, whereas UDP provides connectionless communication

b) ARP

arp

ARP maps IPV4 addresses to MAC addresses on a local network

c) DNS

dns

DNS translates domain names into IP addresses.

d) HTTP

Http

HTTP is used to transfer web resources between clients and servers

e) IP and ICMP

ip, icmp, ipv6, icmpv6

IP handles packet addressing and routing ICMP is used for network diagnostics

f) DHCP

dhcp

DHCP automatically assigns IP addresses to devices. Its four main messages are Discover, offer, Request and ACK

Result:

Network packet were captured and analyzed using Wireshark. Display filters were used to study different protocols